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(a) f (b) $\frac{f}{2}$

(c) $\frac{f}{f+2}$ (d) $\frac{f+2}{f}$

9. If the temperature and pressure of a gas is doubled the mean free path of the gas molecules

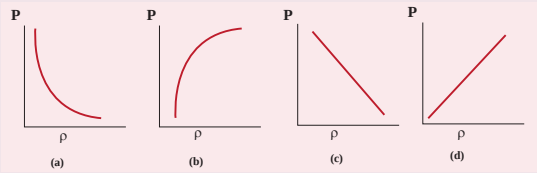
(a) remains same

(b) doubled

(c) tripled

(d) quadrapoled

10. Which of the following shows the correct relationship between the pressure and density of an ideal gas at constant temperature?



11. A sample of gas consists of μ_1 moles of monoatomic molecules, μ_2 moles of diatomic molecules and μ_3 moles of linear triatomic molecules. The gas is kept at high temperature. What is the total number of degrees of freedom?

(a) $[3\mu_1 + 7(\mu_2 + \mu_3)]\, N_A$

(b) $[3\mu_1 + 7\,\mu_2 + 6\mu_3]\, N_A$

(c) $[7\mu_1 + 3(\mu_2 + \mu_3)]\, N_A$

(d) $[3\mu_1 + 6(\mu_2 + \mu_3)]\, N_A$

12. If s_p and s_v denote the specific heats of nitrogen gas per unit mass at constant pressure and constant volume respectively, then (JEE 2007)

(a) $s_p - s_v = 28R$

(b) $s_p - s_v = R/28$

(c) $s_p - s_v = R/14$

(d) $s_p - s_v = R$

13. Which of the following gases will have least rms speed at a given temperature?

(a) Hydrogen

(b) Nitrogen

(c) Oxygen

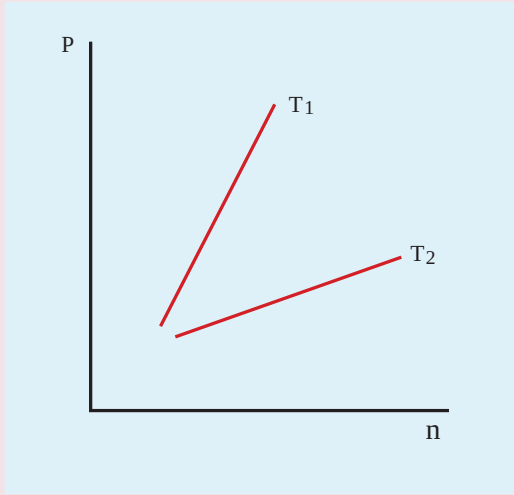
(d) Carbon dioxide

14. For a given gas molecule at a fixed temperature, the area under the Maxwell-Boltzmann distribution curve is equal to

(a) $\frac{PV}{kT}$ (b) $\frac{kT}{PV}$

(c) $\frac{P}{NkT}$ (d) PV

15. The following graph represents the pressure versus number density for ideal gas at two different temperatures T_1 and T_2. The graph implies



(a) $T_1 = T_2$

(b) $T_1 > T_2$

(c) $T_1 < T_2$

d) Cannot be determined

Answers:

1) a	2) c	3) b	4) a
5) a	6) b	7) c	8) d
9) a	10) d	11) a	12) b
13) d	14) a	15) b	